

```
In [2]: !pip install "transformers==4.44.2" "huggingface_hub==0.23.5" --upgrade
#!pip install torch==2.9.1 torchvision==0.24.0 torchaudio==2.9.0 --index-url https:
```

```
Requirement already satisfied: transformers==4.44.2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (4.44.2)
Collecting huggingface_hub==0.23.5
  Using cached huggingface_hub-0.23.5-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: filelock in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (3.19.1)
Requirement already satisfied: numpy>=1.17 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (2.1.2)
Requirement already satisfied: packaging>=20.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (25.0)
Requirement already satisfied: pyyaml>=5.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (2025.11.3)
Requirement already satisfied: requests in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (2.32.5)
Requirement already satisfied: safetensors>=0.4.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (0.7.0)
Requirement already satisfied: tokenizers<0.20,>=0.19 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (0.19.1)
Requirement already satisfied: tqdm>=4.27 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from transformers==4.44.2) (4.67.1)
Requirement already satisfied: fsspec>=2023.5.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface_hub==0.23.5) (2025.9.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface_hub==0.23.5) (4.15.0)
Requirement already satisfied: colorama in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from tqdm>=4.27->transformers==4.44.2) (0.4.6)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests->transformers==4.44.2) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests->transformers==4.44.2) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests->transformers==4.44.2) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests->transformers==4.44.2) (2025.11.12)
Using cached huggingface_hub-0.23.5-py3-none-any.whl (402 kB)
Installing collected packages: huggingface_hub
  Attempting uninstall: huggingface_hub
    Found existing installation: huggingface_hub 1.1.5
    Uninstalling huggingface_hub-1.1.5:
      Successfully uninstalled huggingface_hub-1.1.5
Successfully installed huggingface_hub-0.23.5
```

```
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
datasets 4.4.1 requires huggingface-hub<2.0,>=0.25.0, but you have huggingface-hub 0.23.5 which is incompatible.
```

```
In [4]: !pip install evaluate accelerate
```

Collecting evaluate

Using cached evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)

Collecting accelerate

Downloading accelerate-1.12.0-py3-none-any.whl.metadata (19 kB)

Requirement already satisfied: datasets>=2.0.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (4.4.1)

Requirement already satisfied: numpy>=1.17 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (2.1.2)

Requirement already satisfied: dill in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (0.4.0)

Requirement already satisfied: pandas in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (2.3.3)

Requirement already satisfied: requests>=2.19.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (2.32.5)

Requirement already satisfied: tqdm>=4.62.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (4.67.1)

Requirement already satisfied: xxhash in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (3.6.0)

Requirement already satisfied: multiprocessing in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (0.70.18)

Requirement already satisfied: fsspec>=2021.05.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from fsspec[http]>=2021.05.0->evaluate) (2025.9.0)

Requirement already satisfied: huggingface-hub>=0.7.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (0.23.5)

Requirement already satisfied: packaging in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from evaluate) (25.0)

Requirement already satisfied: psutil in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from accelerate) (7.1.3)

Requirement already satisfied: pyyaml in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from accelerate) (6.0.3)

Requirement already satisfied: torch>=2.0.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from accelerate) (2.9.1+cu128)

Requirement already satisfied: safetensors>=0.4.3 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from accelerate) (0.7.0)

Requirement already satisfied: filelock in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from datasets>=2.0.0->evaluate) (3.19.1)

Requirement already satisfied: pyarrow>=21.0.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from datasets>=2.0.0->evaluate) (22.0.0)

Requirement already satisfied: httpx<1.0.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from datasets>=2.0.0->evaluate) (0.28.1)

Collecting huggingface-hub>=0.7.0 (from evaluate)

Using cached huggingface-hub-1.1.5-py3-none-any.whl.metadata (13 kB)

Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from fsspec[http]>=2021.05.0->evaluate) (3.13.2)

Requirement already satisfied: anyio in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from httpx<1.0.0->datasets>=2.0.0->evaluate) (4.11.0)

Requirement already satisfied: certifi in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from httpx<1.0.0->datasets>=2.0.0->evaluate) (2025.11.12)

Requirement already satisfied: httpcore==1.* in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from httpx<1.0.0->datasets>=2.0.0->evaluate) (1.0.9)

Requirement already satisfied: idna in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from httpx<1.0.0->datasets>=2.0.0->evaluate) (3.11)

Requirement already satisfied: h11>=0.16 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from httpcore==1.*->httpx<1.0.0->datasets>=2.0.0->evaluate) (0.16.0)

Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface-hub>=0.7.0->evaluate) (1.2.0)

Requirement already satisfied: shellingham in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface-hub>=0.7.0->evaluate) (1.5.4)

Requirement already satisfied: typer-slim in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface-hub>=0.7.0->evaluate) (0.20.0)

Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from huggingface-hub>=0.7.0->evaluate) (4.15.0)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (2.6.1)

Requirement already satisfied: aiosignal>=1.4.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.4.0)

Requirement already satisfied: async-timeout<6.0,>=4.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (5.0.1)

Requirement already satisfied: attrs>=17.3.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (25.4.0)

Requirement already satisfied: frozenlist>=1.1.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.8.0)

Requirement already satisfied: multidict<7.0,>=4.5 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (6.7.0)

Requirement already satisfied: propcache>=0.2.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (0.4.1)

Requirement already satisfied: yarll<2.0,>=1.17.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]>=2021.05.0->evaluate) (1.22.0)

Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests>=2.19.0->evaluate) (3.4.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from requests>=2.19.0->evaluate) (2.5.0)

Requirement already satisfied: sympy>=1.13.3 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from torch>=2.0.0->accelerate) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from torch>=2.0.0->accelerate) (3.3)

Requirement already satisfied: jinja2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from torch>=2.0.0->accelerate) (3.1.6)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)

Requirement already satisfied: colorama in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from tqdm>=4.62.1->evaluate) (0.4.6)

Requirement already satisfied: exceptiongroup>=1.0.2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from anyio->httpx<1.0.0->datasets>=2.0.0->evaluate) (1.3.0)

Requirement already satisfied: sniffio>=1.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from anyio->httpx<1.0.0->datasets>=2.0.0->evaluate) (1.3.1)

Requirement already satisfied: MarkupSafe>=2.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from jinja2->torch>=2.0.0->accelerate) (2.1.5)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from pandas->evaluate) (2.9.0.post0)

```

Requirement already satisfied: pytz>=2020.1 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from pandas->evaluate) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from pandas->evaluate) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from python-dateutil>=2.8.2->pandas->evaluate) (1.17.0)
Requirement already satisfied: click>=8.0.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from typer-slim->huggingface-hub>=0.7.0->evaluate) (8.3.1)
Using cached evaluate-0.4.6-py3-none-any.whl (84 kB)
Downloading accelerate-1.12.0-py3-none-any.whl (380 kB)
Using cached huggingface_hub-1.1.5-py3-none-any.whl (516 kB)
Installing collected packages: huggingface-hub, accelerate, evaluate

```

Attempting uninstall: huggingface-hub

Found existing installation: huggingface-hub 0.23.5

Uninstalling huggingface-hub-0.23.5:

Successfully uninstalled huggingface-hub-0.23.5

```

----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 0/3 [huggingface-hub]
----- 1/3 [accelerate]
----- 1/3 [accelerate]
----- 1/3 [accelerate]
----- 1/3 [accelerate]
----- 1/3 [accelerate]
----- 1/3 [accelerate]
----- 2/3 [evaluate]
----- 2/3 [evaluate]
----- 3/3 [evaluate]

```

Successfully installed accelerate-1.12.0 evaluate-0.4.6 huggingface-hub-1.1.5

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

tokenizers 0.19.1 requires huggingface-hub<1.0,>=0.16.4, but you have huggingface-hub 1.1.5 which is incompatible.

transformers 4.44.2 requires huggingface-hub<1.0,>=0.23.2, but you have huggingface-hub 1.1.5 which is incompatible.

In [15]: !pip install scikit-learn

```

Downloading scikit_learn-1.7.2-cp310-cp310-win_amd64.whl.metadata (11 kB)
Requirement already satisfied: numpy>=1.22.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from scikit-learn) (2.1.2)
Requirement already satisfied: scipy>=1.8.0 in c:\users\moham\anaconda3\envs\yolo-normal\lib\site-packages (from scikit-learn) (1.15.3)
Collecting joblib>=1.2.0 (from scikit-learn)
  Using cached joblib-1.5.2-py3-none-any.whl.metadata (5.6 kB)
Collecting threadpoolctl>=3.1.0 (from scikit-learn)
  Using cached threadpoolctl-3.6.0-py3-none-any.whl.metadata (13 kB)
Downloading scikit_learn-1.7.2-cp310-cp310-win_amd64.whl (8.9 MB)
----- 0.0/8.9 MB ? eta -:--:--
----- 8.7/8.9 MB 59.5 MB/s eta 0:00:01
----- 8.9/8.9 MB 45.9 MB/s 0:00:00
Using cached joblib-1.5.2-py3-none-any.whl (308 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Installing collected packages: threadpoolctl, joblib, scikit-learn

```

[illegible]

```

----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
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----- 2/3 [scikit-learn]
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----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 2/3 [scikit-learn]
----- 3/3 [scikit-learn]

```

Successfully installed joblib-1.5.2 scikit-learn-1.7.2 threadpoolctl-3.6.0

```

In [10]: import torch
          print(torch.__version__)
          print(torch.cuda.is_available())
          print(torch.version.cuda)
          print(torch.cuda.get_device_name(0))

```

2.9.1+cu128

True

12.8

NVIDIA GeForce RTX 5060 Laptop GPU

```

In [1]: from datasets import load_dataset
          from transformers import (
              AutoTokenizer,
              AutoModelForSequenceClassification,
              TrainingArguments,
              Trainer,
              DataCollatorWithPadding
          )
          import numpy as np
          import evaluate
          import torch

```

C:\Users\moham\anaconda3\envs\cuda-env\lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidget.s.readthedocs.io/en/stable/user_install.html
 from .autonotebook import tqdm as notebook_tqdm

```

In [2]: dataset = load_dataset("ag_news")

          train_ds = dataset["train"]    # 120,000 samples
          test_ds  = dataset["test"]     # 7,600 samples

```

```
num_labels = 4 # AG News has 4 categories
print(train_ds[0])
```

```
{'text': "Wall St. Bears Claw Back Into the Black (Reuters) Reuters - Short-sellers, Wall Street's dwindling\\band of ultra-cynics, are seeing green again.", 'label': 2}
```

BERT - Model Training

Tokenization and Data Preprocessing

We are doing the following transformations on the data: Now each example has:

- input_ids: [101, 2023, 2003, ...] / after converting each word of the text into id's
- attention_mask: [1, 1, 1, 0, 0, ...] / essentially making only the words of the text to have the attention and paddings added to the text to become unimportant for the classification.
- label: 0-3

```
In [3]: model_name = "bert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)

max_len = 200

def preprocess(batch):
    return tokenizer(
        batch["text"],
        truncation=True,
        padding="max_length", # fixed length = max_len
        max_length=max_len
    )

train_ds_enc = train_ds.map(preprocess, batched=True)
test_ds_enc = test_ds.map(preprocess, batched=True)
```

```
In [4]: train_ds_enc = train_ds_enc.remove_columns(["text"])
test_ds_enc = test_ds_enc.remove_columns(["text"])
```

```
In [5]: train_ds_enc.set_format("torch")
test_ds_enc.set_format("torch")
```

```
In [6]: model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=num_labels
)
```

Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-uncased and are newly initialized: ['classifier.bias', 'classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```
In [7]: accuracy_metric = evaluate.load("accuracy")

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=-1)
    return accuracy_metric.compute(predictions=preds, references=labels)
```

```
In [13]: training_args = TrainingArguments(
    output_dir="./bert_agnews_results",
    learning_rate=2e-5,
    per_device_train_batch_size=6,    # 512 is huge; will likely OOM
    per_device_eval_batch_size=6,
    num_train_epochs=10,
    weight_decay=0.01,
    logging_steps=100,
    fp16=torch.cuda.is_available(),  # only used if GPU & supported
)
```

```
In [14]: data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_ds_enc,
    eval_dataset=test_ds_enc,
    tokenizer=tokenizer,
    data_collator=data_collator,
    compute_metrics=compute_metrics,
)

trainer.train()
```

C:\Users\moahm\AppData\Local\Temp\ipykernel_31748\1537370970.py:4: FutureWarning: `tokenizer` is deprecated and will be removed in version 5.0.0 for `Trainer.\_\_init\_\_`. Use `processing\_class` instead.

```
    trainer = Trainer(
```


[5534/200000 17:48:04 < 625:46:07, 0.09 it/s,

Epoch 0.28/10]

Step	Training Loss
100	0.317900
200	0.301900
300	0.313800
400	0.434800
500	0.443600
600	0.379100
700	0.489300
800	0.452000
900	0.362800
1000	0.416700
1100	0.340600
1200	0.375600
1300	0.299300
1400	0.299400
1500	0.389800
1600	0.361600
1700	0.285300
1800	0.289400
1900	0.404700
2000	0.328300
2100	0.344300
2200	0.381000
2300	0.320000
2400	0.304500
2500	0.257900
2600	0.331600
2700	0.340600
2800	0.286600
2900	0.386100

Step	Training Loss
3000	0.279900
3100	0.332600
3200	0.335600
3300	0.272100
3400	0.306100
3500	0.376200
3600	0.212200
3700	0.298000
3800	0.320700
3900	0.311500
4000	0.277500
4100	0.272500
4200	0.333300
4300	0.323200
4400	0.274600
4500	0.236800
4600	0.260400
4700	0.296300
4800	0.269000
4900	0.252800
5000	0.280200
5100	0.232500
5200	0.341200
5300	0.201400
5400	0.243800
5500	0.301800

KeyboardInterrupt

In []:

In []:

In []: